Cryptography and Embedded System Security
CRAESS_I

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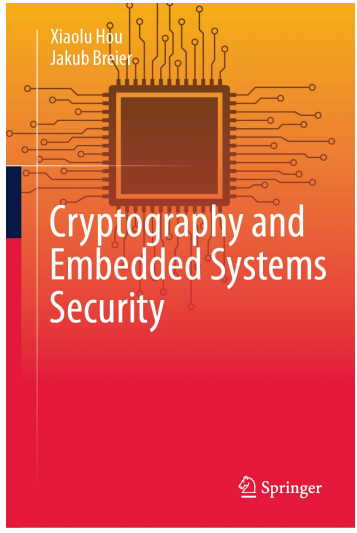
Course Outline

- Abstract algebra and number theory
- Introduction to cryptography
- Symmetric block ciphers and their implementations
- RSA, RSA signatures, and their implementations
- Probability theory and introduction to SCA
- SPA and non-profiled DPA
- Profiled DPA
- SCA countermeasures
- FA on RSA and countermeasures
- FA on symmetric block ciphers
- FA countermeasures for symmetric block cipher
- Practical aspects of physical attacks
  - Invited speaker: Dr. Jakub Breier, Senior security manager, TTControl GmbH

Recommended reading

- Textbook
  - Sections 4.6



Lecture Outline

- Introduction
- Square-and-multiply-always
- Blinding for RSA
- Masking for PRESENT

SCA countermeasures

- Introduction
- Square-and-multiply-always
- Blinding for RSA
- Masking for PRESENT

Countermeasures

- Protocol level: design cryptographic protocols to remain secure under leakage analysis
  - By limiting the number of communications that can be performed with any given key, the attacker can obtain fewer measurements for the same key
- Cryptographic primitive level
  - Proposing new cipher designs that are resistant to side-channel attacks

Countermeasures

- Implementation level
  - Time randomization[1]
  - Encryption of the buses[2]
  - Hiding
  - Masking and blinding

[1] May, D., Muller, H. L., & Smart, N. P. (2001, May). Random register renaming to foil DPA. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 28-38). Springer, Berlin, Heidelberg.

[2] Brier, E., Handschuh, H., & Tymen, C. (2001, May). Fast primitives for internal data scrambling in tamper resistant hardware. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 16-27). Springer, Berlin, Heidelberg.

Countermeasures

- Architecture level
  - Use non-deterministic processor to randomly change the sequence of the executed program during each execution[1]
  - Integrate secure instructions into a non-secure processor[2]

[1] May, D., Muller, H. L., & Smart, N. P. (2001, July). Non-deterministic processors. In Australasian Conference on Information Security and Privacy (pp. 115-129). Springer, Berlin, Heidelberg.

[2] Saputra, H., Vijaykrishnan, N., Kandemir, M., Irwin, M. J., & Brooks, R. (2003). Masking the energy behaviour of encryption algorithms. IEE Proceedings-Computers and Digital Techniques, 150(5), 274-284.

Countermeasures

- Hardware level
  - Conforming glues[1]
  - Protective coating[2]
  - Detachable power supplies[3]

[1] Anderson, R., & Kuhn, M. (1996, November). Tamper resistance – a cautionary note. In Proceedings of the second Usenix workshop on electronic commerce (Vol. 2, pp. 1-11).

[2] Tuyls, P., Schrijen, G. J., Škorić, B., Geloven, J. V., Verhaegh, N., & Wolters, R. (2006, October). Read-proof hardware from protective coatings. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 369-383). Springer, Berlin, Heidelberg.

[3] Shamir, A. (2000, August). Protecting smart cards from passive power analysis with detached power supplies. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 71-77). Springer, Berlin, Heidelberg.

Hiding and masking/blinding

- We have seen how the dependency of a device's leakages (power consumption) on data and operations can be exploited to recover the secret keys of a cryptographic implementation.
- Goal: make the leakage of the DUT independent of the operations being executed and the intermediate values being processed.
- Hiding – reduce or remove the dependence of the leakages on the executed operations or processed data.
  - This can be done by modifying the implementation so that different operations consume a similar amount of energy (balancing the leakages), or so that the leakages are randomized.
- Masking/blinding – remove the dependence of the leakages on the true intermediate values by randomizing the values processed by the DUT.
  - The rationale is that since the value being processed in the DUT is randomized and independent of the intermediate value of the cryptographic computation, we cannot capture information on the actual intermediate value from the leakages.
  - Symmetric block cipher: masking
  - Asymmetric cipher: blinding

Hiding – randomizing power consumption

- Insert random delays (jitter)[1]
- Shuffle the execution order of independent operations
  - Sboxes in AES[2]
  - Randomize the sequence of square and multiply[3]
- Using residue number systems allows us to randomize the representation of modular residues during exponentiation.[4]

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[1]Coron, J. S., & Kizhvatov, I. (2009, September). An efficient method for random delay generation in embedded software. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 156-170). Springer, Berlin, Heidelberg.

[2]Herbst, C., Oswald, E., & Mangard, S. (2006, June). An AES smart card implementation resistant to power analysis attacks. In International conference on applied cryptography and network security. Springer.

[3]Walter, C. D. (2002, February). MIST: An efficient, randomized exponentiation algorithm for resisting power analysis. In Cryptographers' Track at the RSA Conference. Springer.

[4]Bajard, J. C., Imbert, L., Liardet, P. Y., & Teglia, Y. (2004, August). Leak resistant arithmetic. In International Workshop on Cryptographic Hardware and Embedded Systems. Springer.

Hiding – balancing power consumption

- Cell level, logic designs
  - Dual-rail precharge logic (DPL)[1], in the pre-charge phase, the values on the wires are set to a precharge value (either 0 or 1); during the evaluation phase, one wire carries the signal 0 and the other wire carries the signal 1
    - Using code $\{01, 10\}$ for encoding $0, 1$
  - Dynamic and differential logic styles [2]

[1] Tiri, K., & Verbauwhede, I. (2006). A digital design flow for secure integrated circuits. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 25(7), 1197-1208.

[2] Tiri, K., Akmal, M., & Verbauwhede, I. (2002, September). A dynamic and differential CMOS logic with signal independent power consumption to withstand differential power analysis on smart cards. In Proceedings of the 28th European solid-state circuits conference (pp. 403-406). IEEE.

Hiding – balancing power consumption

- Software level
  - DPL in software for symmetric block ciphers[1]
  - DPL in software with proved security for bitsliced implementation of PRESENT[2]
  - Linear complementary dual code[3]
  - Error detecting and correcting code in software[4]
  - Square-and-multiply-always for computing exponentiation in RSA[5]

[1] Hoogvorst, P., Duc, G., & Danger, J. L. (2011). Software implementation of dual-rail representation. COSADE, 51, 24-25.

[2] Rauzy, P., Guilley, S., & Najm, Z. (2013). Formally Proved Security of Assembly Code Against Leakage. IACR Cryptol. ePrint Arch., 2013, 554.

[3] Carlet, C., & Guilley, S. (2015). Complementary dual codes for counter-measures to side-channel attacks. In Coding Theory and Applications (pp. 97-105). Springer, Cham.

[4] Breier, J., & Hou, X. (2017, February). Feeding two cats with one bowl: On designing a fault and side-channel resistant software encoding scheme. In Cryptographers' Track at the RSA Conference (pp. 77-94). Springer, Cham.

[5] Coron, J. S. (1999, August). Resistance against differential power analysis for elliptic curve cryptosystems. In International workshop on cryptographic hardware and embedded systems (pp. 292-302). Springer, Berlin, Heidelberg.

Masking and blinding

- Let v be the secret intermediate value that we would like to mask.
- The masked value, denoted $v_{\mathfrak{m}}$, is obtained by combining v with a random value $\mathfrak{m}$, called a *mask*, using a binary operation $\cdot$ such that

$$v_{\mathfrak{m}} = v \cdot \mathfrak{m}.$$

Masking

- Boolean masking, the binary operation is bitwise `XOR`
- Arithmetic masking, the binary operation is modular addition or modular multiplication
- Affine masking[1]
- Polynomial masking[2]
- Inner product masking[3]

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[1]Willich, M. V. (2001, December). A technique with an information-theoretic basis for protecting secret data from differential power attacks. In IMA International Conference on Cryptography and Coding (pp. 44-62). Springer, Berlin, Heidelberg.

[2]Goubin, L., & Martinelli, A. (2011, September). Protecting AES with Shamir's secret sharing scheme. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 79-94). Springer, Berlin, Heidelberg.

[3]Balasch, J., Faust, S., Gierlichs, B., & Verbauwhede, I. (2012, December). Theory and practice of a leakage resilient masking scheme. In International Conference on the Theory and Application of Cryptology and Information Security (pp. 758-775). Springer, Berlin, Heidelberg.

Masking – more methods in hardware implementations

- Masking buses[1]
- Boolean masking with DPL[2]
- Random precharging[3]

[1]Benini, L., Galati, A., Macii, A., Macii, E., & Poncino, M. (2003, August). Energy-efficient data scrambling on memory-processor interfaces. In Proceedings of the 2003 International Symposium on Low Power Electronics and Design, 2003. ISLPED'03. (pp. 26-29). IEEE.

[2]Popp, T., & Mangard, S. (2005, August). Masked dual-rail pre-charge logic: DPA-resistance without routing constraints. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 172-186). Springer, Berlin, Heidelberg.

[3]Bucci, M., Guglielmo, M., Luzzi, R., & Trifiletti, A. (2004, September). A power consumption randomization countermeasure for DPA-resistant cryptographic processors. In International Workshop on Power and Timing Modeling, Optimization and Simulation (pp. 481-490). Springer, Berlin, Heidelberg.

SCA countermeasures

- Introduction
- Square-and-multiply-always
- Blinding for RSA
- Masking for PRESENT

Motivation

- We have seen an SPA attack on RSA implementations that exploits the part of the square-and-multiply algorithm in which multiplication is carried out only when the secret-key bit is 1.
- A natural countermeasure is to always compute the multiplication, regardless of the value of the secret-key bit.
- Such an algorithm is called the *square-and-multiply-always algorithm*.

Recall – RSA

Definition (RSA)

Let $n = pq$, where p, q are distinct prime numbers. Let $\mathcal{P} = \mathcal{C} = \mathbb{Z}_n$, $\mathcal{K} = \mathbb{Z}^*_{\varphi(n)} - \{ 1 \}$. For any $e \in \mathcal{K}$, define encryption

$$E_e : \mathbb{Z}_n \to \mathbb{Z}_n, \quad m \mapsto m^e \mod n,$$

and the corresponding decryption

$$D_d : \mathbb{Z}_n \to \mathbb{Z}_n, \quad c \mapsto c^d \mod n,$$

where $d = e^{-1} \mod \varphi(n)$. The cryptosystem $(\mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{E}, \mathcal{D})$, where $\mathcal{E} = \{ E_e : e \in \mathcal{K} \}$, $\mathcal{D} = \{ D_d : d \in \mathcal{K} \}$, is called *RSA*.

- $\varphi(n) = (p-1)(q-1)$
- Public key: (n, e), where n is the RSA modulus and e is the encryption exponent
- Private key: d, the decryption exponent

Square-and-multiply algorithm

- Let $n \geq 2$ be an integer, $d \in \mathbb{Z}_{\varphi(n)}$, $a \in \mathbb{Z}_n$
- Binary representation of $d = d_{\ell_d - 1} \dots d_2 d_1 d_0$, where $d_i = 0, 1$ and

$$d = \sum_{i=0}^{\ell_d - 1} d_i 2^i.$$

- We have

$$a^d = a^{\sum_{i=0}^{\ell_d - 1} d_i 2^i} = \prod_{i=0}^{\ell_d - 1} (a^{2^i})^{d_i} = \prod_{0 \leq i < \ell_d, d_i = 1} a^{2^i}.$$

- Thus, to compute $a^d \mod n$, we can
  - First compute a^{2^i} for $0 \leq i < \ell_d$
  - Then a^d is a product of a^{2^i} for which $d_i = 1$
- Compared to $d - 1$ modular multiplications, this requires at most $2\ell_d$ modular multiplications, where ℓ_d is the bit length of d.

Square-and-multiply algorithms

Algorithm 1: Right-to-left

Input: $n,\ a,\ d$// $n \in \mathbb{Z}, n \geq 2;\ a \in \mathbb{Z}_n;$ $d \in \mathbb{Z}_{\varphi(n)}$ has bit length ℓ_d

Output: $a^d \mod n$

1 result $= 1$, $t = a$
2 **for** $i = 0$, $i < \ell_d$, $i++$ **do**
  // ith bit of d is 1
3 **if** $d_i = 1$ **then**
    // multiply by a^{2^i}
4 result $=$ result $* t \mod n$// $a^d = \prod_{0 \leq i < \ell_d, d_i = 1} a^{2^i}$
  // $t = a^{2^{i+1}}$
5 $t = t * t \mod n$
6 **return** result

Algorithm 2: Left-to-right

Input: $n,\ a,\ d$// $n \in \mathbb{Z}, n \geq 2;\ a \in \mathbb{Z}_n;$ $d \in \mathbb{Z}_{\varphi(n)}$

Output: $a^d \mod n$

1 $t = 1$
2 **for** $i = \ell_d - 1$, $i \geq 0$, $i--$ **do**
3 $t = t * t \mod n$
  // ith bit of d is 1
4 **if** $d_i = 1$ **then**
5 $t = a * t \mod n$
6 **return** t

Right-to-left square-and-multiply-always

Input: $n,\ a,\ d$// $n \in \mathbb{Z}, n \geq 2;\ a \in \mathbb{Z}_n;$ $d \in \mathbb{Z}_{\varphi(n)}$ has bit length ℓ_d

Output: $a^d \mod n$

1 result $= 1$, $t = a$

2 **for** $i = 0$, $i < \ell_d$, $i++$ **do**

  // ith bit of d is 1

3 **if** $d_i = 1$ **then**

    // multiply by a^{2^i}

4 result $=$ result $* t \mod n$// $a^d = \prod_{0 \leq i < \ell_d, d_i = 1} a^{2^i}$

  // $t = a^{2^{i+1}}$

5 $t = t * t \mod n$

6 **return** result

Algorithm 3: Square-and-multiply-always

Input: $n,\ a,\ d$

Output: $a^d \mod n$

1 result $= 1$, $t = a$

2 **for** $i = 0$, $i < \ell_d$, $i++$ **do**

  // ith bit of d is 1

3 **if** $d_i = 1$ **then**

    // multiply by a^{2^i}

4 result $=$ result $* t \mod n$

5 **else**

    // compute multiplication and discard the result

6 tmp $=$ result $* t \mod n$

  // $t = a^{2^{i+1}}$

7 $t = t * t \mod n$

8 **return** result

Left-to-right square-and-multiply-always

Input: $n,\ a,\ d$// $n \in \mathbb{Z}, n \geq 2;\ a \in \mathbb{Z}_n;$ $d \in \mathbb{Z}_{\varphi(n)}$ has bit length ℓ_d
Output: $a^d \mod n$

```
1 t = 1
2 for i = ℓ_d − 1, i ≥ 0, i − − do
3     t = t ∗ t mod n
      // ith bit of d is 1
4     if d_i = 1 then
5         t = a ∗ t mod n
6 return t
```

Algorithm 4: Square-and-multiply-always

Input: $n,\ a,\ d$
Output: $a^d \mod n$

```
1 t = 1
2 for i = ℓ_d − 1, i ≥ 0, i − − do
3     t = t ∗ t mod n
      // ith bit of d is 1
4     if d_i = 1 then
5         t = a ∗ t mod n
6     else
          // ith bit of d is 0, compute
             multiplication and discard the
             result
7         tmp = a ∗ t mod n
8 return t
```

SPA on left-to-right square-and-multiply algorithm

For our experiment, we have set

$$p = 29, \quad q = 41, \quad n = 1189, \quad \varphi(n) = 1120, \quad e = 3, \quad d = 747$$

Algorithm 5: Left-to-right square-and-multiply algorithm for computing modular exponentiation with parameters from above.

Input: a// $a \in \mathbb{Z}_{1189}$
Output: $a^{747} \mod 1189$

```
1 n = 1189
2 dbin = [1,1,0,1,0,1,1,1,0,1]// binary representation of d = 747, d_0 = 1, d_1 = 1
3 ℓ_d = length of dbin// bit length of d
4 t = 1
5 for i = ℓ_d − 1, i ≥ 0, i − − do
6     t = t ∗ t mod n
      // ith bit of d is 1
7     if d_i = 1 then
8         t = a ∗ t mod n
9 return t
```

SPA on left-to-right square-and-multiply algorithm

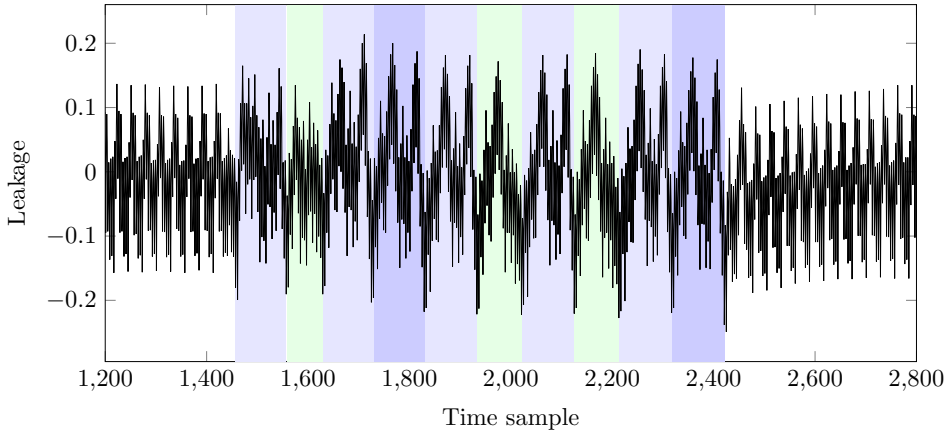
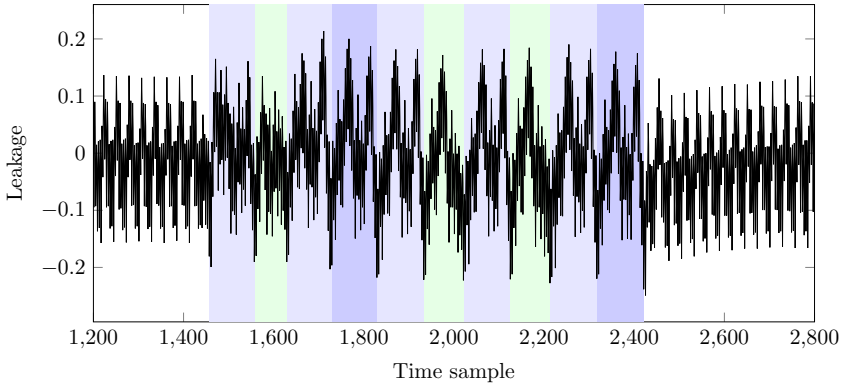


Figure: Green patterns (single peak cluster) $\rightarrow d_i = 0$; blue patterns (multiple peak clusters) $\rightarrow d_i = 1$

We can then read out the values of the bits d_i for $i = \ell_d - 1, \ldots, 0$.

SPA on left-to-right square-and-multiply algorithm



We can then read out the values of the bits d_i for $i = \ell_d - 1, \ldots, 0$.

$$1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1.$$

Finally, we recover the secret key

$$d = 1011101011_2 = 747.$$

Implementation with countermeasure

$$p = 29, \quad q = 41, \quad n = 1189, \quad \varphi(n) = 1120, \quad e = 3, \quad d = 747$$

Algorithm 6: Left-to-right square-and-multiply-always algorithm.

Input: a// $a \in \mathbb{Z}_{1189}$
Output: $a^{747} \mod 1189$

```
1  n = 1189
2  dbin = [1,1,0,1,0,1,1,1,0,1]// binary representation of d = 747, d_0 = 1, d_1 = 1
3  ℓ_d = length of dbin// bit length of d
4  t = 1
5  for i = ℓ_d − 1, i ≥ 0, i − − do
6      t = t ∗ t mod n
7      if d_i = 1 then
8          t = a ∗ t mod n
9      else
           // ith bit of d is 0, compute multiplication and discard the result
10         tmp = a ∗ t mod n
11 return t
```

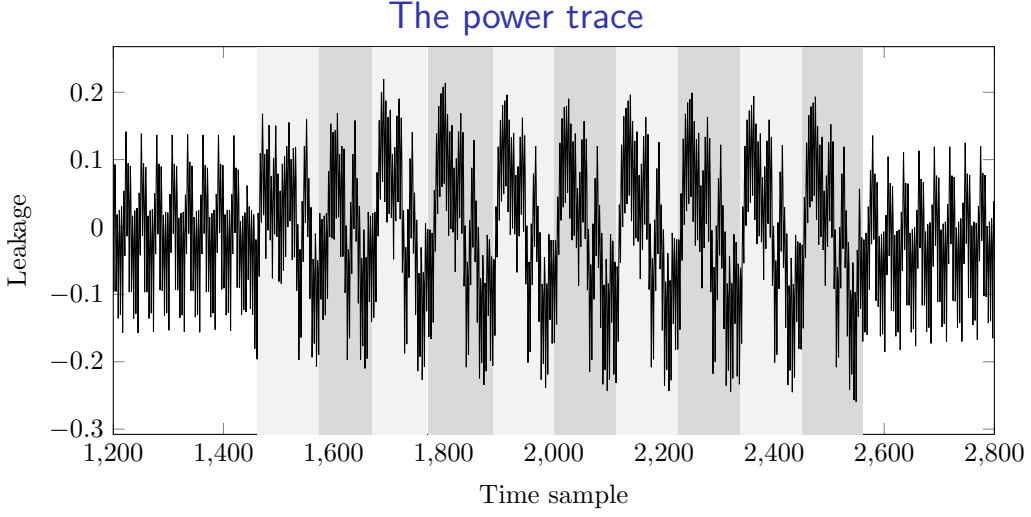
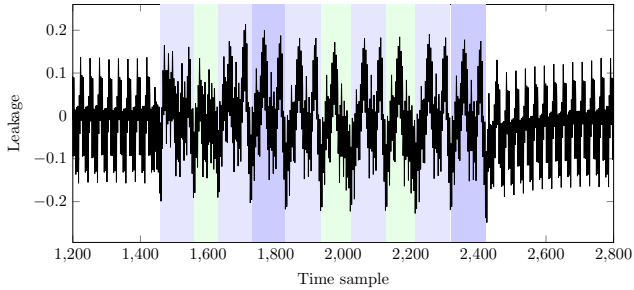
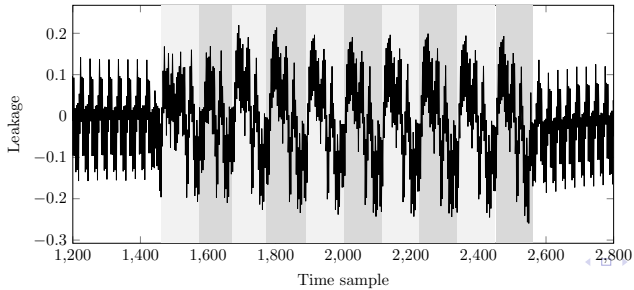


Figure: One trace corresponding to the computation of left-to-right square-and-multiply-always algorithm. We can see ten similar patterns. But in this case, all of them have more than one peak cluster.

Comparison


Leakage
0.2
0.1
0
−0.1
−0.2
1,200
1,400
1,600
1,800
2,000
2,200
2,400
2,600
2,800
Time sample


Leakage
0.2
0.1
0
−0.1
−0.2
−0.3
1,200
1,400
1,600
1,800
2,000
2,200
2,400
2,600
2,800
Time sample

SCA countermeasures

- Introduction
- Square-and-multiply-always
- Blinding for RSA
- Masking for PRESENT

Some history

- The application of arithmetic masks in the context of public-key cryptosystems is called blinding
- First suggested by Kocher[1]
- Later formalized by J.-S. Coron[2]
- There is also a patent related to blinding methods[3]

[1] Kocher, P. C. (1996, August). Timing attacks on implementations of Diffie-Hellman, RSA, DSS, and other systems. In Annual International Cryptology Conference (pp. 104-113). Springer, Berlin, Heidelberg.

[2] Coron, J. S. (1999, August). Resistance against differential power analysis for elliptic curve cryptosystems. In International workshop on cryptographic hardware and embedded systems (pp. 292-302). Springer, Berlin, Heidelberg.

[3] Kocher, P. C., & Jaffe, J. M. (2001). U.S. Patent No. 6,298,442. Washington, DC: U.S. Patent and Trademark Office.

Notations

- Let p, q be two distinct odd primes, $n = pq$ be the RSA modulus
- $d \in \mathbb{Z}^*_{\varphi(n)}$: the private key for RSA
- $e = d^{-1} \mod \varphi(n)$: public key for RSA
- The SPA attack we have discussed exploits leakages during the computation of

$$a^d \mod n$$

  for some $a \in \mathbb{Z}_n$.
- The attack can be carried out during RSA signature generation or RSA decryption.
- DPA attacks can also be applied to RSA implementations

Blinding

- Types of blinding: exponent blinding, message blinding, and modulus blinding
- These countermeasures are mainly designed against DPA attacks
- It is recommended to blind secret values during the computation.
- The masks and blinded values should be updated frequently, or even during the computation.
- In this case, it becomes more difficult for the attacker to combine partial information obtained from previously blinded values with newly leaked information.

Exponent blinding

- Generate a random number $\lambda \in [0, 2^\ell - 1]$
- Typically, for RSA modulus of bit length 1024, we take $\ell = 20$ or 30 to guarantee a reasonable overhead
- Instead of computing

$$a^d \mod n,$$

- We compute

$$a^{d+\lambda\varphi(n)} \mod n.$$

Exponent blinding

- Instead of computing

$$a^d \mod n,$$

- We compute

$$a^{d+\lambda\varphi(n)} \mod n.$$

Theorem

Let p and q be two distinct primes and $n = pq$. For any $a \in \mathbb{Z}$, any positive integer b, and any nonnegative integer λ, we have

$$a^{b+\lambda\varphi(n)} \equiv a^b \mod n.$$

In particular, if $b > 0$ and $b \not\equiv 0 \mod \varphi(n)$, then

$$a^b \equiv a^{b \mod \varphi(n)} \mod n.$$

Example for exponent blinding

Example

$$p = 3, \quad q = 5, \quad n = 15, \quad \varphi(n) = 8, \quad e = 3, \quad d = 3$$

Take $a = 8$ and $\lambda = 2$.

$$a^d \mod n = 8^3 \mod 15 = 2.$$

With the countermeasure, we have

$$\begin{aligned} a^{d+\lambda\varphi(n)} \mod n &= 8^{3+2\times 8} \mod 15 = 8^{19} \mod 15 = 8 \times (8^2)^9 \mod 15 \\ &= 8 \times 4^9 \mod 15 = 8 \times 4 \times (4^2)^4 \mod 15 = 32 \mod 15 = 2. \end{aligned}$$

Effectiveness against our SPA attack

- With one decryption computation, even if we recover the bits of the exponent, it will be $d+$ a random number
- But if we attack two decryption computations, we obtain

$$d + \lambda_1\varphi(n), \quad d + \lambda_2\varphi(n).$$

- Their difference is

$$(\lambda_1 - \lambda_2)\varphi(n),$$

  so we obtain a multiple of $\varphi(n)$ rather than $\varphi(n)$ itself.
- Effective against certain DPA attacks

Message blinding

Take a random number λ such that $\gcd(\lambda, n) = 1$, compute

$$a_1 = \lambda^e \mod n, \quad a_2 = \lambda^{-1} \mod n.$$

And to get

$$a^d \mod n,$$

we calculate

$$(((aa_1)^d \mod n)a_2) \mod n.$$

Message blinding

- Since

$$ed \equiv 1 \mod \varphi(n)$$

and $\gcd(\lambda, n) = 1$, same theorem as before gives

$$\lambda^{ed} \equiv \lambda \mod n \Longrightarrow \lambda^{ed-1} \equiv 1 \mod n.$$

- Then

$$\begin{aligned} (((aa_1)^d \mod n)a_2) \mod n &= (((a\lambda^e \mod n)^d \mod n)(\lambda^{-1} \mod n)) \mod n \\ &= ((a^d \mod n)(\lambda^{ed-1} \mod n)) \mod n = a^d \mod n \end{aligned}$$

- The first mask a_1 randomizes the input of the computation
- The second mask a_2 corrects the output to the expected result.

Example for message blinding

$$a_1 = \lambda^e \mod n, \quad a_2 = \lambda^{-1} \mod n, \quad a^d \mod n = (((aa_1)^d \mod n)a_2) \mod n.$$

Example

$$p = 3, \quad q = 5, \quad n = 15, \quad e = 3, \quad d = 3, \quad a = 8, \quad \varphi(n) = 8$$

Take $\lambda = 4$, which is coprime with n. Then

$$a_1 = \lambda^e \mod n = 4^3 \mod 15 = 64 \mod 15 = 4.$$

By the extended Euclidean algorithm

$$15 = 4 \times 3 + 3, \quad 4 = 3 + 1 \Longrightarrow 1 = 4 - 3 = 4 - (15 - 4 \times 3) = 4 \times 4 - 15$$

and

$$a_2 = \lambda^{-1} \mod n = 4.$$

Example for message blinding

$$a_1 = \lambda^e \bmod n, \quad a_2 = \lambda^{-1} \bmod n, \quad a^d \bmod n = (((aa_1)^d \bmod n)a_2) \bmod n.$$

Example

$$p = 3, \quad q = 5, \quad n = 15, \quad e = 3, \quad d = 3, \quad a = 8, \quad \varphi(n) = 8$$

Take $\lambda = 4$, which is coprime with n. Then

$$a_1 = 4, \quad a_2 = 4.$$

$$\begin{aligned} (((aa_1)^d \bmod n)a_2) \bmod n &= (((8 \times 4)^3 \bmod 15) \times 4) \bmod 15 \\ &= ((2^3 \bmod 15) \times 4) \bmod 15 = 32 \bmod 15 = 2. \end{aligned}$$

We can compute

$$a^d \bmod n = 8^3 \bmod 15 = 512 \bmod 15 = 2.$$

Modulus blinding

Generate a random number λ and compute

$$(a^d \mod (\lambda n)) \mod n.$$

It is easy to see that

$$(a^d \mod (\lambda n)) \mod n = a^d \mod n.$$

Example for modulus blinding

Example

$$p = 3, \quad q = 5, \quad n = 15, \quad e = 3, \quad d = 3, \quad a = 8, \quad \varphi(n) = 8$$

Take $\lambda = 4$, then

$$\begin{aligned} (a^d \bmod (\lambda n)) \bmod n &= (8^3 \bmod (4 \times 15)) \bmod 15 \\ &= (512 \bmod 60) \bmod 15 = 32 \bmod 15 = 2. \end{aligned}$$

We can check that

$$a^d \bmod n = 2.$$

Effectiveness on DPA attacks

- DPA attacks that rely on knowing certain intermediate values related to a cannot be carried out when a or n is randomized

Attacks on Countermeasures

- Fouque, P. A., & Valette, F. (2003, September). The doubling attack–why upwards is better than downwards. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 269-280). Springer, Berlin, Heidelberg.
  - Recover a blinded secret exponent by SPA
  - Only works for left to right implementation for square-and-multiply algorithm
- Witteman, M. F., van Woudenberg, J. G., & Menarini, F. (2011, February). Defeating RSA multiply-always and message blinding countermeasures. In Cryptographers' Track at the RSA Conference (pp. 77-88). Springer, Berlin, Heidelberg.
  - DPA attack
- Fouque, P. A., Réal, D., Valette, F., & Drissi, M. (2008, August). The carry leakage on the randomized exponent countermeasure. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 198-213). Springer, Berlin, Heidelberg.
  - Exploit leakage during the computation of the random exponent

SCA countermeasures

- Introduction
- Square-and-multiply-always
- Blinding for RSA
- Masking for PRESENT

Boolean masking

- Can achieve provable security under suitable leakage assumptions, provided that the source of randomness is truly random[1]
- The cryptographic algorithm needs to be changed a bit for us to carry out computations with the masked intermediate values and keep track of all the masks
- At the end of the encryption, we can remove the masks to output the original ciphertext

[1] Prouff, E., & Rivain, M. (2013, May). Masking against side-channel attacks: A formal security proof. In Annual International Conference on the Theory and Applications of Cryptographic Techniques (pp. 142-159). Springer, Berlin, Heidelberg.

Masking scheme

- A *masking scheme* specifies how masks are applied to the plaintext and intermediate values, as well as how they are removed from the ciphertext.
- There are a few principles we follow for a masking scheme design
  - All intermediate values should be masked during the computation. In particular, we would apply masks to the plaintext (and the key).
  - We assume the attacker does not have knowledge of the masks – otherwise, the attacker can carry out a DPA attack by making hypotheses about the key values.
  - When some intermediate values are to be XOR-ed with each other (e.g. in the AES MixColumns operation), different masks should be applied to each of them.
  - Each encryption has a different set of randomly generated masks.
- For any function f, the mask that is applied to an input of f is called the *input mask* of f.
- The corresponding mask for the output is called the *output mask* for f.

Linear function

Definition

Let $f : \mathbb{F}_2^{m_1} \to \mathbb{F}_2^{m_2}$ be a function, where m_1 and m_2 are positive integers. f is said to be *linear* (w.r.t. $\oplus$) if for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{F}_2^{m_1}$, we have

$$f(\boldsymbol{x} \oplus \boldsymbol{y}) = f(\boldsymbol{x}) \oplus f(\boldsymbol{y}).$$

f is *non-linear* if it is not linear.

For functions of several inputs, we use the same terminology: a map $g : \mathbb{F}_2^{m_1} \times \cdots \times \mathbb{F}_2^{m_r} \to \mathbb{F}_2^{m}$ is called linear if it is linear with respect to the componentwise operation $\oplus$ on its inputs.

Example

- AddRoundKey operation in AES, viewed as the two-input map $(\boldsymbol{x}, \boldsymbol{k}) \mapsto \boldsymbol{x} \oplus \boldsymbol{k}$, is linear. More generally, bitwise `XOR` is linear as a function of its two inputs.
- DES Sboxes are non-linear functions. In symmetric block ciphers, Sboxes are designed to be non-linear.

Boolean masking for linear functions

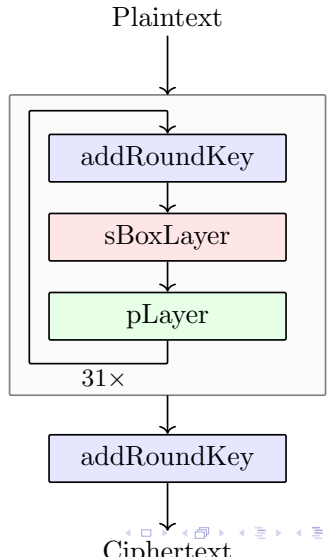
- With Boolean masking, it is easy to keep track of the masks with linear operations.
- Let f be a linear operation and take any input of f, $\boldsymbol{v}$, with a corresponding mask $\mathfrak{m}$, we have
$$f(\boldsymbol{v} \oplus \mathfrak{m}) = f(\boldsymbol{v}) \oplus f(\mathfrak{m}).$$
- When the input mask is $\mathfrak{m}$, the output mask is given by $f(\mathfrak{m})$.
- One of the main challenges in designing a masking scheme is to find ways to keep track of masks for non-linear operations.

PRESENT – encryption

- Block length: 64
- Round function: addRoundKey, sBoxLayer, and pLayer.
- After 31 rounds, addRoundKey is applied again before the ciphertext output
- Key length: 80 or 128.

Remark

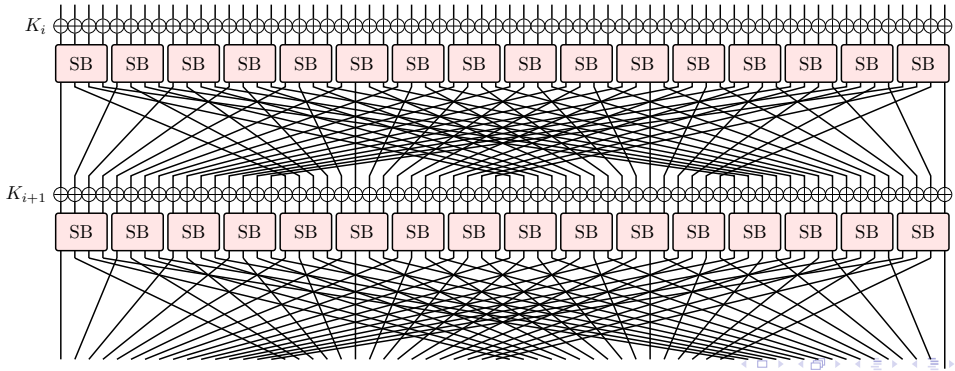
For PRESENT specification, we consider the 0th bit of a value as the rightmost bit in its binary representation. For example, the 0th bit of $3 = 011_2$ is 1, the 1st bit is 1 and the 2nd bit is 0.



PRESENT – sBoxLayer

- sBoxLayer applies sixteen 4-bit Sboxes, one to each nibble of the current cipher state.
- For example, if the input is `0`, the output is `C`.

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
C	5	6	B	9	0	A	D	3	E	F	8	4	7	1	2



PRESENT – pLayer

pLayer permutes the 64 bits using the following formula:

$$\text{pLayer}(j) = \left\lfloor \frac{j}{4} \right\rfloor + (j \mod 4) \times 16,$$

where j denotes the bit position.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	16	32	48	1	17	33	49	2	18	34	50	3	19	35	51
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
4	20	36	52	5	21	37	53	6	22	38	54	7	23	39	55
32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
8	24	40	56	9	25	41	57	10	26	42	58	11	27	43	59
48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
12	28	44	60	13	29	45	61	14	30	46	62	15	31	47	63

Masking PRESENT Sbox

- Compute a lookup table T1 such that for any $\boldsymbol{v} \in \mathbb{F}_2^4$, any input mask $\mathfrak{m}_{\mathsf{in}}$ and its corresponding output mask $\mathfrak{m}_{\mathsf{out}}$ for PRESENT Sbox,

$$\mathsf{T1}[\boldsymbol{v} \oplus \mathfrak{m}_{\mathsf{in}}, \mathfrak{m}_{\mathsf{in}}] = \mathsf{SB}(\boldsymbol{v}) \oplus \mathfrak{m}_{\mathsf{out}}.$$

- Table T2 helps us keep track of the masks

$$\mathsf{T2}[\mathfrak{m}_{\mathsf{in}}] = \mathfrak{m}_{\mathsf{out}}, \quad \mathfrak{m}_{\mathsf{in}} = 0, 1, \ldots, \mathtt{F}.$$

- Do not need to generate a masked Sbox lookup table whenever the input mask for the Sbox changes.
- T1 has an 8-bit input and a 4-bit output, so the storage required is $2^8 \times 4 = 2^{10}$ bits, or 2^7 bytes.
- The table T2 requires $2^4 \times 4 = 64$ bits of memory.

Masked cipher state

- Since the pLayer operation is linear, we can simply apply pLayer to the masks to keep track of their changes.
- We represent the intermediate values of PRESENT encryption as

$$\boldsymbol{b}_{15}, \boldsymbol{b}_{14}, \ldots, \boldsymbol{b}_1, \boldsymbol{b}_0,$$

  where each $\boldsymbol{b}_j$ denotes a nibble of the cipher state.
- At the beginning of one encryption, we randomly generate 16 masks, and each is applied to one nibble of the plaintext.
- Suppose the cipher state at the input of round i is of the following format:

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}_{15,\mathsf{in}}^{i-1}, \boldsymbol{b}_{14} \oplus \mathfrak{m}_{14,\mathsf{in}}^{i-1}, \ldots, \boldsymbol{b}_1 \oplus \mathfrak{m}_{1,\mathsf{in}}^{i-1}, \boldsymbol{b}_0 \oplus \mathfrak{m}_{0,\mathsf{in}}^{i-1}.$$

Masked addRoundKey

Suppose the cipher state at the input of round i is of the following format:

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}_{15,\mathsf{in}}^{i-1}, \boldsymbol{b}_{14} \oplus \mathfrak{m}_{14,\mathsf{in}}^{i-1}, \ldots, \boldsymbol{b}_{1} \oplus \mathfrak{m}_{1,\mathsf{in}}^{i-1}, \boldsymbol{b}_{0} \oplus \mathfrak{m}_{0,\mathsf{in}}^{i-1}.$$

- We do not apply masks to the round keys. Consequently, after the addRoundKey operation, each nibble of the cipher state still has the same mask

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}_{15,\mathsf{in}}^{i-1}, \boldsymbol{b}_{14} \oplus \mathfrak{m}_{14,\mathsf{in}}^{i-1}, \ldots, \boldsymbol{b}_{1} \oplus \mathfrak{m}_{1,\mathsf{in}}^{i-1}, \boldsymbol{b}_{0} \oplus \mathfrak{m}_{0,\mathsf{in}}^{i-1},$$

Masked sBoxLayer

Let

$$\mathfrak{m}_{j,\mathsf{out}}^{i-1} = \mathsf{T2}\left[\mathfrak{m}_{j,\mathsf{in}}^{i-1}\right], \quad j = 0, 1, \ldots, 15,$$

denote the output mask for PRESENT Sbox corresponding to the input mask $\mathfrak{m}_{j,\mathsf{in}}^{i-1}$. Then after sBoxLayer, the cipher state is as follows:

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}_{15,\mathsf{out}}^{i-1}, \boldsymbol{b}_{14} \oplus \mathfrak{m}_{14,\mathsf{out}}^{i-1}, \ldots, \boldsymbol{b}_{1} \oplus \mathfrak{m}_{1,\mathsf{out}}^{i-1}, \boldsymbol{b}_{0} \oplus \mathfrak{m}_{0,\mathsf{out}}^{i-1},$$

Masked pLayer

- We apply the pLayer operation to both the cipher state and the mask for the whole cipher state, i.e. the string obtained by concatenating all 16 masks $\mathfrak{m}_{j,\text{out}}^{i-1}$:

$$\mathfrak{m}_{15,\text{out}}^{i-1}, \mathfrak{m}_{14,\text{out}}^{i-1}, \ldots, \mathfrak{m}_{1,\text{out}}^{i-1}, \mathfrak{m}_{0,\text{out}}^{i-1}.$$

- After pLayer, masks for each nibble of the cipher state will be changed and the cipher state will become:

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}_{15,\text{in}}^{i}, \boldsymbol{b}_{14} \oplus \mathfrak{m}_{14,\text{in}}^{i}, \ldots, \boldsymbol{b}_{1} \oplus \mathfrak{m}_{1,\text{in}}^{i}, \boldsymbol{b}_{0} \oplus \mathfrak{m}_{0,\text{in}}^{i}.$$

- Where

$$\mathfrak{m}_{15,\text{in}}^{i}, \mathfrak{m}_{14,\text{in}}^{i}, \ldots, \mathfrak{m}_{1,\text{in}}^{i}, \mathfrak{m}_{0,\text{in}}^{i} = \text{pLayer}(\mathfrak{m}_{15,\text{out}}^{i-1}, \mathfrak{m}_{14,\text{out}}^{i-1}, \ldots, \mathfrak{m}_{1,\text{out}}^{i-1}, \mathfrak{m}_{0,\text{out}}^{i-1}).$$

- Consequently, $\mathfrak{m}_{j,\text{in}}^{i}$ will be the input mask for the jth Sbox in round $i+1$.

Final addRoundKey

- Finally, after 31 rounds, we have another addRoundKey operation, which does not change the masks of the cipher state.
- The cipher state will be

$$\boldsymbol{b}_{15} \oplus \mathfrak{m}^{31}_{15,\mathsf{in}}, \boldsymbol{b}_{14} \oplus \mathfrak{m}^{31}_{14,\mathsf{in}}, \ldots, \boldsymbol{b}_{1} \oplus \mathfrak{m}^{31}_{1,\mathsf{in}}, \boldsymbol{b}_{0} \oplus \mathfrak{m}^{31}_{0,\mathsf{in}}.$$

- To get the unmasked ciphertext, we remove the masks by XORing the cipher state with

$$\mathfrak{m}^{31}_{15,\mathsf{in}}, \mathfrak{m}^{31}_{14,\mathsf{in}}, \ldots, \mathfrak{m}^{31}_{1,\mathsf{in}}, \mathfrak{m}^{31}_{0,\mathsf{in}}.$$

Algorithmic description

- Input: p, T1, T2, K_i $(i = 1, 2, \ldots, 32)$
- Output: ciphertext

1 randomly generate 16 masks $\mathtt{m}_0, \mathtt{m}_1, \ldots, \mathtt{m}_{15}$
2 **array** of size 16 state $= p \oplus \mathtt{m}_{15}, \mathtt{m}_{14}, \ldots, \mathtt{m}_1, \mathtt{m}_0$// mask the jth nibble of the plaintext with $\mathtt{m}_j$, each entry of the array is one masked nibble
3 **array** of size 16 masks $= \mathtt{m}_{15}, \mathtt{m}_{14}, \ldots, \mathtt{m}_1, \mathtt{m}_0$
4 **for** $i = 1$, $i \leq 31$, $i++$ **do**
5 state $=$ addRoundKey(state, K_i)
6 **for** $j = 0$, $j < 16$, $j++$ **do**
// for each nibble
7 state$[j]$ $=$ T1[state$[j]$, masks$[j]$]// masked Sbox computation
8 masks$[j]$ $=$ T2[masks$[j]$]// record the output masks of Sbox computation
9 state $=$ pLayer(state)// apply pLayer to the cipher state
10 masks $=$ pLayer(masks)// apply pLayer to the masks
11 state $=$ addRoundKey(state, K_{32})
12 state $=$ state $\oplus$ masks// remove mask from the ciphertext
13 **return** state

Design of T2

- It is suggested that T2 be designed such that all possible values of $\mathfrak{m}_{\text{in}} \oplus \mathfrak{m}_{\text{out}}$ appear.
- For example, one possible choice of T2 is as follows

$\mathfrak{m}_{\text{in,SB}}$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$\mathfrak{m}_{\text{out,SB}} = \text{T2}[\mathfrak{m}_{\text{in,SB}}]$	E	4	F	9	0	3	D	5	7	8	A	2	B	1	6	C
$\mathfrak{m}_{\text{in,SB}} \oplus \mathfrak{m}_{\text{out,SB}}$	E	5	D	A	4	6	B	2	F	1	0	9	7	C	8	3

Sasdrich, P., Bock, R., & Moradi, A. (2018). Threshold Implementation in Software: Case Study of PRESENT. In Constructive Side-Channel Analysis and Secure Design: 9th International Workshop, COSADE 2018, Singapore, April 23–24, 2018, Proceedings 9 (pp. 227-244). Springer International Publishing.

Design of T2

- In fact, in general, we have the following observations:
- Let f be any function and let $\mathfrak{m}_{\mathsf{in},f}$ (resp. $\mathfrak{m}_{\mathsf{out},f}$) denote its input mask (resp. output mask).
- For any input $\boldsymbol{x}$ of f, we have

$$(\boldsymbol{x} \oplus f(\boldsymbol{x})) \oplus (\mathfrak{m}_{\mathsf{in},f} \oplus \mathfrak{m}_{\mathsf{out},f}) = (\boldsymbol{x} \oplus \mathfrak{m}_{\mathsf{in},f}) \oplus (f(\boldsymbol{x}) \oplus \mathfrak{m}_{\mathsf{out},f}).$$

- Thus, when choosing the input mask $\mathfrak{m}_{\mathsf{in},f}$ and output mask $\mathfrak{m}_{\mathsf{out},f}$ of f, we need to ensure that all possible values of $\mathfrak{m}_{\mathsf{in},f} \oplus \mathfrak{m}_{\mathsf{out},f}$ appear.
- Otherwise, the distribution induced by $(\boldsymbol{x} \oplus f(\boldsymbol{x})) \oplus (\mathfrak{m}_{\mathsf{in},f} \oplus \mathfrak{m}_{\mathsf{out},f})$ will not be uniform, and the signal corresponding to the value of $\boldsymbol{x} \oplus f(\boldsymbol{x})$ cannot be properly concealed, making it vulnerable to DPA attacks.

Higher-order masking

- The masked value v_m is related to the original value v and the mask $\mathfrak{m}$ through a binary operation $v_{\mathfrak{m}} = v \cdot \mathfrak{m}$.
- In the language of *secret sharing*[1], we can say that the secret value v is represented by two *shares*, $v_{\mathfrak{m}}$ and $\mathfrak{m}$.
- Given only one of the two shares, no information about v can be revealed.
- Instead of two shares (or one mask), we can also use several shares, resulting in a higher-order masking.
- In particular, a d*th-order masking scheme* applies $d - 1$ masks to the secret value v.

[1] Beimel, A. (2011, May). Secret-sharing schemes: A survey. In International conference on coding and cryptology (pp. 11-46). Berlin, Heidelberg: Springer Berlin Heidelberg.

Higher order DPA

- The DPA attack we have discussed uses one intermediate value – such a DPA attack is also called *first-order DPA*.
- When information leakage from several intermediate values (e.g. by combining leakages from a few time samples) is analyzed, we have a *higher-order DPA*[1]
- In general, the number of traces required for a higher-order DPA grows quickly with the noise level. Under standard assumptions, the complexity increases significantly with the masking order.[2]

[1] Chari, S., Jutla, C. S., Rao, J. R., & Rohatgi, P. (1999, August). Towards sound approaches to counteract power-analysis attacks. In Annual International Cryptology Conference (pp. 398-412). Springer, Berlin, Heidelberg.

[2] Prouff, E., & Rivain, M. (2013, May). Masking against side-channel attacks: A formal security proof. In Annual International Conference on the Theory and Applications of Cryptographic Techniques (pp. 142-159). Springer, Berlin, Heidelberg.

Security of Boolean masking

- Let us take a secret value $\boldsymbol{v}$ of bit length at most m_v.
- The masked value is given by $\boldsymbol{v} \oplus \mathfrak{m}$, where $\mathfrak{m} \in \mathbb{F}_2^{m_v}$.
- We can consider the value of $\mathfrak{m}$ as a discrete random variable.
- In case the distribution induced by this random variable is uniform on $\mathbb{F}_2^{m_v}$, the distribution induced by the value of $\boldsymbol{v} \oplus \mathfrak{m}$ is also uniform on $\mathbb{F}_2^{m_v}$ regardless of the value of $\boldsymbol{v}$.
- Thus, we expect the leakage to be independent of $\boldsymbol{v}$ when only first-order DPA is carried out.
- For Boolean masking schemes with one mask, security proofs have been given[1]
- Security proofs for higher-order masking have also been given[2]

[1] Blömer, J., Guajardo, J., & Krummel, V. (2004, August). Provably secure masking of AES. In International workshop on selected areas in cryptography (pp. 69-83). Springer, Berlin, Heidelberg.

[2] Rivain, M., & Prouff, E. (2010, August). Provably secure higher-order masking of AES. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 413-427). Springer, Berlin, Heidelberg.

Further Reading

- Masking was first proposed by Goubin and Patarin[1] and Chari et al.[2] independently
- Hardware implementations of masking are vulnerable to DPA attacks due to glitches in CMOS circuits[3]
- Threshold Implementation, which is based on multiparty computation, [4], was introduced as a principled way to realize Boolean masking on hardware platforms[5]

[1] Goubin, L., & Patarin, J. (1999, August). DES and differential power analysis the "Duplication" method. In International Workshop on Cryptographic Hardware and Embedded Systems (pp. 158-172). Springer, Berlin, Heidelberg.

[2] Chari, S., Jutla, C. S., Rao, J. R., & Rohatgi, P. (1999, August). Towards sound approaches to counteract power-analysis attacks. In Annual International Cryptology Conference (pp. 398-412). Springer, Berlin, Heidelberg.

[3] Mangard, S., Popp, T., & Gammel, B. M. (2005, February). Side-channel leakage of masked CMOS gates. In Cryptographers' Track at the RSA Conference. Springer, Berlin, Heidelberg.

[4] Cramer, R., & Damgård, I. (2005). Multiparty computation, an introduction. In Contemporary cryptology (pp. 41-87). Birkhäuser Basel.

[5] Nikova, S., Rijmen, V., & Schläffer, M. (2011). Secure hardware implementation of nonlinear functions in the presence of glitches. Journal of Cryptology, 24(2), 292-321.